



Annamalai University

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
FACULTY OF ENGINEERING AND TECHNOLOGY, ANNAMALAI UNIVERSITY, ANNAMALAINAGAR - 608002
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PROJECT

Automated Wire Harness Management and Diagnostic System

Guide :

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SUBMITTED BY B.E (ECE) -VIII SEMESTER

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Abstract

- The project develops an automated wire harness diagnostic system to address limitations of traditional manual testing methods.
- It integrates ESP32 microcontroller for signal integrity, continuity testing, and propagation delay measurement.
- Arduino Nano module performs electrical load testing and thermal analysis using ACS712 current sensor and DS18B20 temperature sensor.
- The system successfully detects multiple faults including open/short circuits, high resistance, signal delay, and overheating.
- Real-time visual and audio feedback improves testing efficiency, accuracy, reliability, and safety in automotive and industrial applications.

Objectives

- To design and develop an automated wire harness diagnostic system for efficient testing.
- To integrate signal integrity and continuity testing using ESP32 microcontroller.
- To perform electrical load and thermal analysis using Arduino Nano, ACS712, and DS18B20 sensors.
- To detect faults such as open circuit, short circuit, high resistance, propagation delay, and overheating.
- To provide real-time results with visual and audio feedback for improved reliability and safety.

Problem Statement

- Traditional wire harness testing is mostly manual, time-consuming, and prone to human error.
- Manual methods cannot effectively detect subtle faults like signal propagation delay and gradual overheating under load.
- Increasing complexity of modern wire harnesses in automotive and industrial systems demands faster and more reliable testing.
- Existing systems often focus only on continuity testing and lack integrated thermal and load analysis.
- Poor wire harness quality can lead to system failures, safety hazards, and high maintenance costs.

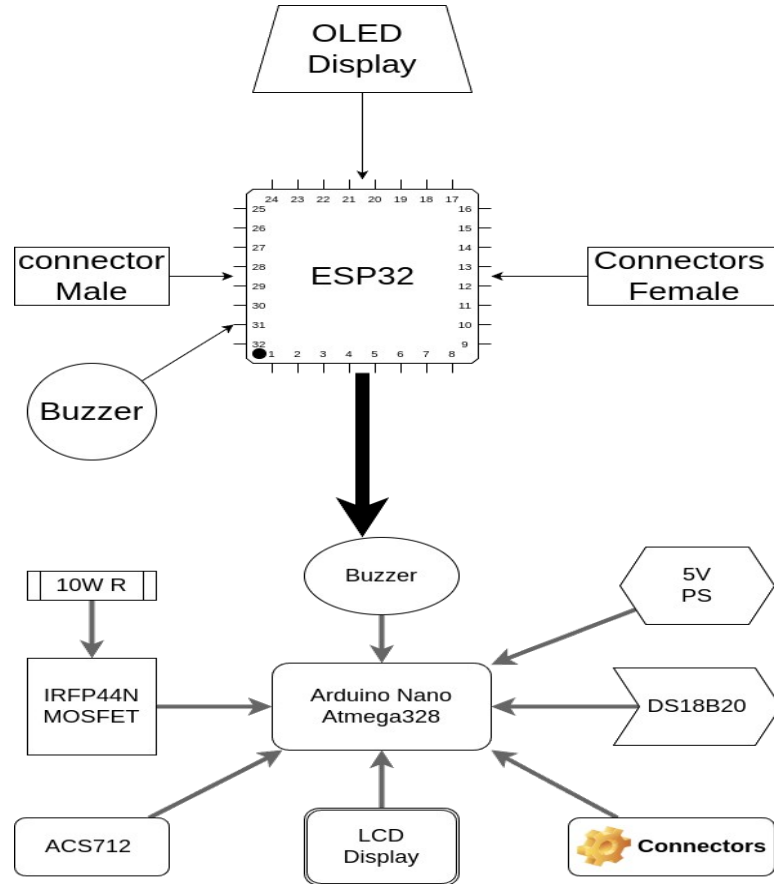
Literature Review

Authors	Title	Content
J. Smith & R. Brown	Wire Harness Systems (2018)	Emphasized design & maintenance for reliability; faults can cause system failure
P. Singh & A. Rao	Integrated Diagnostics (2023)	Multi-parameter (signal + current + thermal) testing is essential
R. Gupta & P. Sharma	Automated Testing (2022)	Microcontroller systems improve efficiency but lack thermal analysis

Hardware and Software

- ESP32-WROOM Microcontroller
 - Arduino Nano (ATmega328)
 - ACS712 Current Sensor (30A)
 - DS18B20 Temperature Sensor
 - IRFP150N MOSFET
 - 10W Power Resistor
 - OLED Display (SSD1306)
 - 16x2 LCD Display with I2C
 - Battery
- Thonny
 - Arduino

Block Diagram



Methodology

- Develops an automated diagnostic system for wire harness testing
- Combines two independent modules for comprehensive analysis
- Module 1: Signal integrity & continuity testing (ESP32)
- Module 2: Electrical load & thermal analysis (Arduino Nano)
- Aims to detect electrical, signal, and thermal faults in real-time

Methodology

- Signal Testing Unit: ESP32-WROOM + OLED Display + Buzzer
- Performs continuity and propagation delay testing
- Load & Thermal Testing Unit: Arduino Nano + ACS712 + DS18B20
- Applies controlled load and monitors current & temperature
- Uses male/female connectors for easy wire harness interfacing
- Provides visual (OLED + LCD)

Methodology

- Initialize both microcontrollers and peripherals
- Stage 1: ESP32 performs signal integrity test (one wire at a time)
- Measures propagation delay → PASS / WARNING / FAIL
- Stage 2: Arduino applies load via MOSFET & 10W resistor
- Monitors current (ACS712) and temperature (DS18B20)
- Real-time display of results with buzzer alerts for faults
- Complete evaluation of electrical and thermal performance

Results and Conclusion

- The developed system successfully detects open circuits, short circuits, high resistance connections, and propagation delays with high accuracy.
- Real-time load and thermal analysis effectively identifies overheating and overcurrent conditions under electrical stress.
- The dual-module approach (ESP32 + Arduino Nano) provides reliable visual and audio feedback for quick fault diagnosis.
- The automated system significantly reduces testing time and human error compared to traditional manual methods.
- The project demonstrates an efficient, cost-effective solution that enhances the reliability and safety of wire harnesses in automotive and industrial applications.

